

MENG 4205.002**LAB 9: IMPACT-CHARPY AND IZOD**

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ABSTRACT

The goals of this lab were to determine the impact characteristics of several engineering materials through Charpy and Izod testing and to define the mechanical friction losses of the impact testing pendulum device and what it means for operation. Impact testing is a form of destructive test used to define the toughness or energy absorbed to failure of a material. The EEICI impact testing device uses a pendulum from which it drops from a predetermined height and strikes a notched specimen; by determining the change in degree of the pendulum from before strike to after failure, absorbed energy to failure can be determined. This is achieved through a Charpy test and an Izod test where the orientation and geometry of the strike are different. Additionally, a friction loss test was performed to evaluate the mechanical losses of the system independent of failure, bringing greater reliability to absorbed energy values. These results provide an understanding of which of the three materials tested possesses greater toughness and highlight the need for impact testing devices to be subject to mechanical loss corrections.

Keywords: Charpy impact test; Izod impact test; pendulum energy; fracture toughness; absorbed energy; dynamic loading; notch sensitivity; pendulum friction losses; impact resistance; mechanical properties

Nomenclature

m	Mass of pendulum (kg)
g	Acceleration due to gravity (m/s^2)
L	Effective pendulum length (m)
α	Initial pendulum angle before impact ($^\circ$)
β	Final pendulum angle after impact ($^\circ$)
$E_{initial_pow}$	Initial potential energy of pendulum (J)
E_{final_pow}	Remaining potential energy after fracture (J)
$E_{absorbed}$	Energy absorbed during impact (J)
$E_{friction}$	Energy lost due to frictional effects (J)
$E_{indicator}$	Additional energy losses associated with pointer mechanism (J)

h	Height corresponding to pendulum angle (m)
Notch	Reduced cross-sectional area to initiate crack propagation (m)

1. INTRODUCTION

Impact resistance is a mechanical property required by materials when applied loads may occur unexpectedly - such as shock, vibration, etc. [1]. While strength is measured under static loading conditions (i.e., static strength tests), impact properties determine the ability of a material to resist energy absorption conditions (on quickly applied deformation). In this way, impact testing provides an index of toughness and vulnerability for brittle fracture. The two standardized approaches to impact testing are Charpy and Izod tests. Both break a notched specimen with a pendulum but differ in orientation, clamping, and point of impact (not to mention microscopic differences). The Charpy notches and breaks the specimen in the horizontal position with both ends supported, while the Izod applies a vertical cantilevered specimen that's clamped at one end and struck at the free edge. Such arrangements cause different stress conditions through the specimen so that different values for absorbed energy are presented.

Impact testing relates to a pendulum capable of striking the specimen relative to a known position (angle) to measure impact of toughness. The new position (angle) reached by the pendulum after absorbing energy via fracture is the final position (angle). Thus, the work by gravity (energy) calculated from the two measurements of initial position and final position is the impact of toughness. However, since mechanical systems operate through other mechanical systems - essentially pendulums will always experience some friction at bearings through use and air resistance, for example, but rarely any measured for data adjustment) thus an additional friction-loss test was assessed during this lab, relative to Charpy and Izod impact properties which explore the integrity of energy transfer, fracture behaviors, and energy corrections for mechanical systems [2].

The objective of this lab was to investigate the impact toughness properties of various materials using Charpy and Izod

impact testing methods as well as investigate friction losses inherent to the pendulum system; the energy results collected are then compared to determine which material works best and a friction-loss correction will provide a better measurement for any of the specimens tested.

2. THEORY

Impact testing is based on energy conservation principles [3]. When the pendulum is raised to an initial angle α , gravitational potential energy is stored according to:

$$E_{initial_pow} = m g L (1 - \cos \alpha) \quad (1)$$

where m is the pendulum mass, g is gravitational acceleration, and L is the pendulum's length. After striking the specimen, the pendulum rises to a smaller angle β , corresponding to a remaining potential energy:

$$E_{final_pow} = m g L (1 - \cos \beta) \quad (2)$$

The absorbed energy impact is the difference:

$$E_{absorbed} = m g L (\cos \beta - \cos \alpha) \quad (3)$$

This absorbed energy represents the toughness of the material [4]. Due to geometric and clamping differences, Charpy and Izod tests do not produce identical toughness values [1][2].

2.1 Izod Method

In the Izod configuration *Figure 1*, the sample is secured in a cantilever position vertically and the hammer strikes at a predetermined height above the clamp. Therefore, this supports configuration creates a different stress pattern and generally lower absorbed energy.

2.2 Charpy Method

Charpy configuration utilizes the notched sample in a horizontal position, supported at both ends and the hammer strikes perpendicular to the notch. The Charpy method generally yields higher absorbed energy values due to this support condition being more evenly distributed.

2.3 Pendulum Friction Losses

To compensate for impact energy readings, the losses due to friction experienced by the pendulum must be taken into consideration. To measure system losses, a pendulum is released without a sample and the difference from start height to end height is noted. Then, a second release is performed, incorporating the .080 kg added mass into the equation to measure indicator losses. These are the changes that will compensate for mechanical losses other than those incurred during impact with a sample.

3. METHODS

3.1 Preparation

Charpy, Izod and friction-loss tests were performed on the EEICI impact testing machine. The pendulum is set at 0.330 m

height, there are Charpy/Izod interchangeable hammers, a protractor and angle gauge to assess initial (where the pendulum strikes the sample) and final (where the pendulum comes to rest) angles of movement as well as vices which hold the samples in the vertical (Izod) position or horizontal (Charpy) position. In addition, there are extra weights that are adjustable and part of the assembly for the various impact energies. The impact Charpy energies were assessed as 5J, 7.5J and 10J and the impact Izod energy was assessed as 8.5J [2]. The samples provided for testing are 3D printed PLA, PETG and Onyx. The samples have a standard geometry notch for ease of crack propagation. The sample was placed in the vice which was hand tightened with a standard hex key. All testing was done on a level machine using a leveling bubble. The friction loss test was done by basically just letting the pendulum fall without a sample in place to assess how much energy is lost in the machine itself.

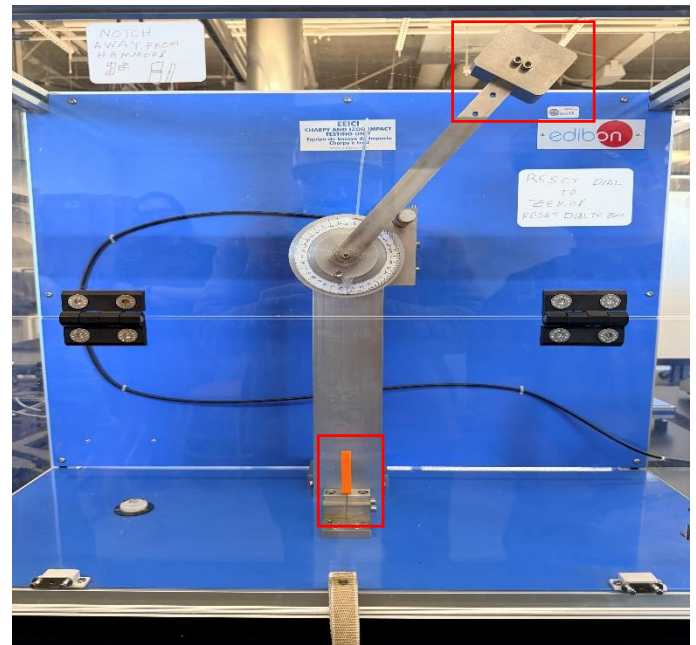


Figure 1 Orange PLA under the Izod Method

3.2 Data Collection Procedure

To prepare for all EEICI testing, the EEICI unit was leveled with its built-in bubble level, establishing a constant direction of force for the pendulum. Then, the pendulum was set in its highest position of 124° to be a predetermined initial position for each testing run. For Charpy, the sample was placed in a horizontal position in the Charpy vice with the notch facing opposite the hammer. An Allen key tightened each clamp screw in an effort to avoid material pre-stresses. The hammer configuration needed was assessed for the necessary amount of impact energy, and the pendulum was adjusted and released to shatter the sample. The

final angle was assessed with the pointer and scale on the protractor.

For Izod, the same configuration was assessed, albeit the sample was placed in a vertical position within the Izod vice. The hammer impacted the sample 0.022 m above the sample clamp point, per manual standards. Notations were made of all initial angles and final angles of all samples tested.

For those impacts that needed to only assess friction losses, no sample was in place. Thus, the same process of achieving the 124° setting and releasing it was done without any counter force/energy absorbed. The angle was recorded post-impact. In addition, to assess losses from indicators, 0.080 kg was strung from the pendulum frame, and the same process was repeated. These values will be accounted as friction values to the absorbed energy values for Charpy and Izod tests.

3.3 Data Reduction and Calculations

For each sample, the necessary experimental information was the initial release angle of the pendulum and the final angle post-collision. The energy absorbed was determined by the difference in gravitational potential energy from the initial height to the final height based upon the known mass of the pendulum and effective length. Experimental friction losses were determined separately and subtracted from absorbed energy values to generate corrected toughness values. Use the tables below to enter your measured values.

4. EXPERIMENTS

4.1 Results



Figure 2 Charpy fracture appearance of PLA-based specimens tested in the horizontal configuration



Figure 3 Izod fracture appearance of PLA-based specimens tested in the vertical cantilever configuration

Table 1 Calibration Parameters for the Impact Testing Unit

Parameter	Value
Initial Pendulum Angle α'	124
Final Reference Angle β'	135.5
Impact height above grips	22 mm

Table 2 Final Angles and Absorbed Energy for PLA Materials in Charpy an Izod Testing

Material	Charpy β	Charpy (J)	Izod β	Izod (J)
Onyx	81	1.97	71	4.08
Orange PLA	86	1.73	91	2.50
PG PLA	105	0.83	90	2.62
Black PLA	104	0.87	103	1.58

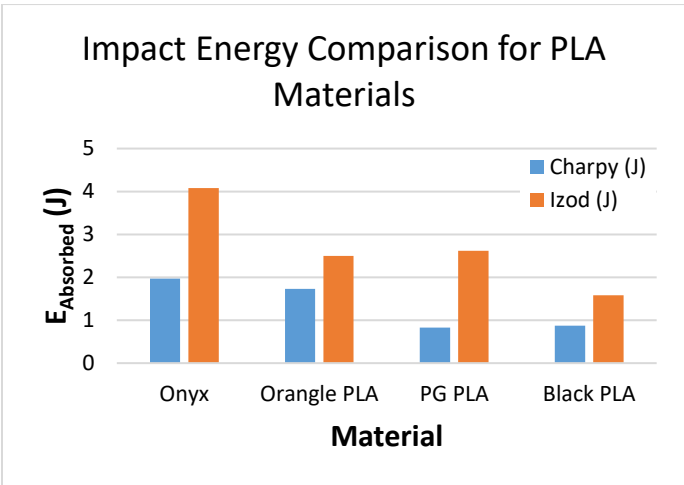


Figure 4 Impact Energy Comparison for PLA Materials under Charpy and Izod Testing

Table 1 contains calibration values for the pendulum experiment with respect to release angle and angle of marker reference. Table 2 marks the angles at final position of pendulum with impact energy absorbed by Charpy and Izod for each specimen. Figures 2 and 3 present the Charpy and Izod fracture surfaces of each material, respectively, a qualitative analysis that indicates a mechanical distinction in fracture pattern between loading types. Figure 4 details the absorbed energy comparison which states that each material absorbed more energy tested via Izod than Charpy.

4.2 Sample Calculations

Below is a sample calculation for determining the absorbed impact energy for a Charpy specimen.

1. Converting angle to height

The height corresponding to a pendulum angle is determined from:

$$h = L (1 - \cos \theta) \quad (4)$$

Assume the following sample data:

Pendulum length, $L = 0.330$ m

Mass, $m = 1.5$ kg

Initial angle, $\alpha = 150^\circ$

Final angle, $\beta = 40^\circ$

Convert angles to height:

Initial height:

$$h_{initial} = 0.330 (1 - \cos 150^\circ)$$

$$h_{initial} = 0.330 (1 - (-0.8660))$$

$$h_{initial} = 0.330 (1 + 0.8660)$$

$$h_{initial} = 0.330 (1.8660)$$

$$h_{initial} \approx 0.6158 \text{ m}$$

Final height:

$$h_{final} = 0.330 (1 - \cos 40^\circ)$$

$$h_{final} = 0.330 (1 - 0.7660)$$

$$h_{final} = 0.330 (0.2340)$$

$$h_{final} \approx 0.0772 \text{ m}$$

2. Calculating initial and final energies

$$E_{initial_pow} = m g h_{initial}$$

$$E_{initial_pow} = (1.5)(9.81)(0.6158)$$

$$E_{initial_pow} \approx 9.05 \text{ J}$$

$$E_{final_pow} = m g h_{final}$$

$$E_{final_pow} = (1.5)(9.81)(0.0772)$$

$$E_{final_pow} \approx 1.14 \text{ J}$$

3. Absorbed impact energy

$$E_{absorbed} = E_{initial_pow} - E_{final_pow}$$

$$E_{absorbed} = 9.05 \text{ J} - 1.14 \text{ J}$$

$$E_{absorbed} = 7.91 \text{ J}$$

This represents the true absorbed impact energy of the specimen.

4.3 Discussion

The results from the experiments showed that there was a statistically significant difference for impact toughness between both materials and loading orientations. According to Table 2 and Figure 4, all materials exhibit higher levels of energy absorbed in the Izod configuration versus Charpy configuration. This corroborates expectations as the Izod configuration loads the sample in a cantilevered position which creates a greater bending moment at the notch and requires additional energy to initiate fracture and sustain subsequent fracture propagation.

Fractures in Figures 2 and 3 support the results seen in Table 2. More ductile materials (Onyx) have more random fracture planes due to excess energy absorption, while PG PLA and Black PLA fractures are clean with minor deformations at 90° perpendicular to the notch plane exhibiting their more brittle characteristics. Additionally, Orange PLA, the middle ground, has both attributes of ductile fractures. Thus, these confirming visuals line up with the quantifiable energy values and their relation to impact fracture performance is due to material characteristics.

Any outlier absorbed energies may be due to sample orientation, printing differences or slight discrepancies of notching sharpness or measurement timing. However, for the most part, these findings correlate with anticipated dynamic fracture performance for these materials.

5. Conclusion

The EEICI pendulum was also used to perform Charpy and Izod impact tests. The values for absorbed energy obtained from angle difference measurements from start to finish of pendulum travel can be used comparatively as relative values of toughness for the materials tested. Notably, the absorbed energy value for the Izod test was greater than that of Charpy due to the impact settings of a cantilever vs. more stable (lever) impact. Onyx yielded the most toughness, while PG PLA and Black PLA were

the least tough, which was confirmed by the post-test fracture surface appearance seen in *Figures 2* and *3*. Ultimately, this test reinforced concepts learned in material science thus far related to dynamic loading, stress raisers and fracture mechanics, but the geometry of the specimen tested and each materials composition impacted the level of impact loading stresses each could withstand.

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